CHAPTER

12 Principles of oncology

Learning objectives

To understand:
- The biological nature of cancer
- That curative treatment is only one component in the overall management of cancer
- The principles of cancer prevention, early detection and screening
- The principles underlying non-surgical treatments for cancer

To appreciate:
- The principles of cancer aetiology and the major causative factors
- The multidisciplinary management of cancer
- The distinction between palliative care and end-of-life care
- The principles of palliative care

WHAT IS CANCER?

History

The word 'cancer' is credited to Hippocrates (460 BCE–370 BCE), who is widely agreed to be the father of medicine, and comes from the Greek word for a crab, referring to the finger-like projections of a cancer from a central mass, which have similarities to a crab's claws and legs.

The study of cancer has long been a part of clinical medicine: theories have moved from divine intervention and are now firmly based on the molecular origins of cancer. Rudolf Virchow was the first to demonstrate that cancer is a disease of cells and that the disease progresses as a result of abnormal proliferation, encapsulated by his dictum *omnes cellula e cellula* (every cell from a cell). In 1914, Theodor Boveri pointed out the importance of chromosomal abnormalities in cancer cells and, in the 1940s, Oswald Avery demonstrated that DNA was the genetic material within the chromosomes. In 1953, Watson and Crick described the structure of DNA, which was the key discovery leading to the understanding of the molecular biology of cancer. This understanding has allowed the investigation and understanding of the molecular mechanisms whereby cancer cells are formed and their abnormal behaviours are mediated; in turn, this has allowed modern molecular-based therapies to be developed.

The hallmarks of cancer

Cancer cells are able to proliferate in an uncontrolled fashion; their ability to divide and spread is unbounded. Cancer cell growth destroys first the tissue from which they arise and eventually the person in which they are present.

In order to survive, divide, invade and spread, cancer cells have to acquire a number of characteristics. No one characteristic is sufficient and not all characteristics are absolutely necessary. These features, based on articles by Hanahan and Weinberg, are given in ***Summary box 12.1***.

Establish an autonomous lineage

Cells develop independence from the normal signals that control supply and demand. The healing of a wound is a physiological process; the cellular response is exquisitely coordinated so that proliferation occurs when it is needed and ceases when it is no longer required. The whole process is controlled by a series of

Hippocrates, 460 BCE–375 BCE, was a Greek Physician and, by common consent, 'the father of medicine'.
Rudolf Ludwig Carl Virchow, 1821–1902, Professor of Pathology, Berlin, Germany.
Theodor Heinrich Boveri, 1862–1913, Professor of Zoology and Comparative Anatomy, Würzburg, Germany.
Oswald Theodore Avery, 1877–1955, bacteriologist, Rockefeller Institute, New York, NY, USA.
James Dewey Watson, b.1928, American biologist who worked in Cambridge, UK, and later became Director of the Cold Spring Harbor Laboratory, New York, NY, USA.
Francis Harry Compton Crick, 1916–2004, British molecular biologist who worked at the Cavendish Laboratory, Cambridge, UK, and later at the Salk Institute, San Diego, CA, USA. Watson and Crick shared the 1962 Nobel Prize in Physiology or Medicine with ***Maurice Hugh Frederick Wilkins***, 1916–2004, of Kings College, London, UK.
Douglas Hanahan, b.1951, American biologist and director of the Swiss Institute for Experimental Cancer Research (ISREC), Lausanne, Switzerland.
Robert Allan Weinberg, b.1942, The Whitehead Institute of Biomedical Research and Department of Biology, The Massachusetts Institute of Technology, Cambridge, MA, USA.

Summary box 12.1

Features of malignant transformation

- Establish an autonomous lineage
  - Resist signals that inhibit growth
  - Sustain proliferative signalling
- Obtain replicative immortality
- Evade apoptosis
- Acquire angiogenic competence
- Acquire ability to invade, disseminate and implant
- Evocation of inflammation
- Evade detection/elimination
- Loss of specialist cell function
- Develop ability to change energy metabolism

signals telling cells when, and when not, to divide. Cancer cells escape from this normal system of checks and balances: they grow and proliferate in the absence of external stimuli and regardless of signals telling them to desist.

Oncogenes are genes with the potential to cause cancer if mutated and expressed at high levels; they are key factors in carcinogenesis. Most oncogenes are normally involved in physiological processes, i.e. cell growth, but if mutated they can predispose a cell to cancer and in concert with other oncogenes can enable cancer cell survival and development of an established tumour. The implication is that we all carry the seeds of our own destruction: genetic sequences that, through mutation, can turn into active oncogenes and thereby cause malignant transformation. Indeed, through study of the timing of key genetic events in adult cancer development, a handful of cells develop the founder mutations of cancer development several decades prior to diagnosis. If the individual is unfortunate enough to accumulate further mutations in key driver genes, that cell becomes malignant and, if it proliferates, a cancer may develop. Only very rarely is a single mutation sufficient to cause cancer; multiple mutations are usually required. Colorectal cancer provides the classical example of how multiple mutations are necessary for the complete transformation from normal cell to malignant cell. Vogelstein and his colleagues identified the genes implicated and also postulated that it is necessary to have mutations in all the relevant genes; they also noted that these mutations must be acquired in a specific sequence for malignant transformation to occur.

Obtain replicative immortality

According to the Hayflick hypothesis, normal cells are permitted to undergo only a finite number of divisions. For humans this number is between 40 and 60. The limitation is imposed by the progressive shortening of the end of the chromosome (the telomere) that occurs each time a cell divides; eventually the lineage will die out. Cancer cells utilise the enzyme telomerase to rebuild the telomere at each cell division, such that there is no telomeric shortening, and the lineage will never die out. The cancer cell hence develops immortality.

Evade apoptosis

Apoptosis, taken from the Greek for 'leaf fall', is a form of programmed cell death that occurs as the direct result of internal cellular events instructing the cell to die. Unlike necrosis, which is a form of traumatic cell death resulting from acute cellular injury, apoptosis is an orderly and internally driven process. The cell dismantles itself neatly for disposal (***Figure 12.1***). There is minimal inflammatory response. Apoptosis is a physiological process. Cells that are redundant normally die by apoptosis and this is an important self-regulatory mechanism in growth and development, i.e. cells in the web space of the embryo die by apoptosis, or lymphocytes that could react to self. Genes, such as *p53*, that can activate apoptosis function as tumour suppressor genes. Mutation in such genes causes a loss of this inhibitory function, which will contribute to malignant transformation as apoptosis is evaded; this means that the wrong cells can be in the wrong places at the wrong times.

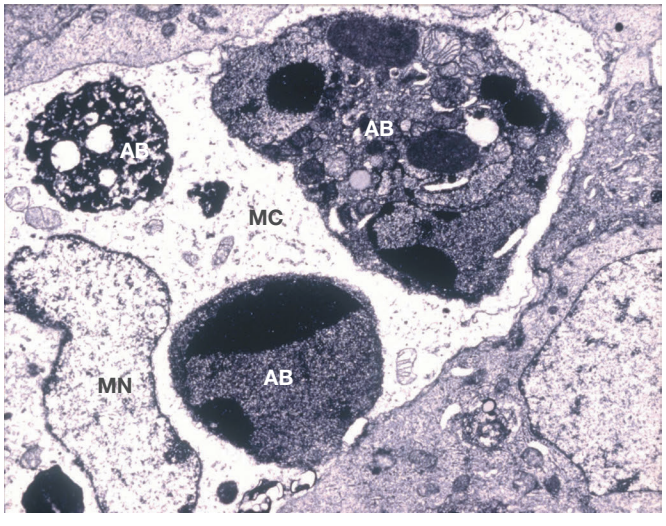


Figure 12.1 Electron micrograph of apoptotic bodies (AB) engulfed by a macrophage. Note the macrophage nucleus (MN) and macrophage cytoplasm (MC).

Acquire angiogenic competence

A mass of cancer cells cannot, in the absence of a blood supply, grow beyond a diameter of about 1 mm. This places a severe restriction on the capabilities of the tumour (note that the word tumour means swelling and does not mean the lesion is malignant, although 'tumour' is often taken by patients to be synonymous with cancer). It cannot grow much larger or spread widely within the body. If, however, the mass of cancer cells is able to attract or to construct a blood supply then it is able to quit its dormant state and behave in a far

Bert Vogelstein, b.1949, molecular biologist, Johns Hopkins Hospital, Baltimore, MD, USA.
Leonard Hayflick, b.1928, while working at the Wistar Institute in Philadelphia in 1962, he noted that normal mammalian cells growing in culture had a limited, rather than an indefinite, capacity for self-replication.

more aggressive fashion. The ability of a cancer to form blood vessels is termed angiogenesis and is a key feature of malignant transformation.

Acquire ability to invade

Cancer cells have no respect for the structure of normal tissues. They acquire the ability to breach the basement membrane and gain direct access to blood and lymph vessels. Cancer cells use three main mechanisms to facilitate invasion: (i) cause a rise in the interstitial pressure within a tissue; (ii) secrete enzymes that dissolve extracellular matrix; and (iii) become mobile. Unrestrained proliferation and a lack of contact inhibition enable cancer cells to exert pressure directly on the surrounding tissue and push beyond the normal limits. They secrete collagenases and proteases that chemically dissolve any extracellular boundaries that would otherwise limit their spread through tissues and, by modulating the expression of cell surface molecules called integrins, are able to detach themselves from the extracellular matrix. The abnormal integrins associated with malignancy can also transmit signals from the environment to the cytoplasm and nucleus of the cancer cells ('outside-in signalling') and these signals can induce increased motility. These processes are similar to those involved in normal development, i.e. in the migration of the neural crest or the formation of the heart. Epithelial cells behave as if they were mesenchymal cells and the process is termed epithelial–mesenchymal transition (EMT). EMT is a crucial step in malignant transformation and many of the genes and proteins implicated in the formation of cancer control processes are involved in EMT, e.g. *Src*, *Ras*, integrins, *Wnt*/β-catenin, Notch.

Acquire ability to disseminate and implant

Once cancer cells gain access to vascular and lymphovascular spaces, they can be readily distributed systemically throughout the body. This is not, of itself, sufficient to cause tumours to develop at distant sites. The cells also need to acquire the ability to implant. As Paget pointed out over a century ago, there is a crucial relationship here between the seed (the tumour cell) and the soil (the distant tissue). Most of the cancer cells discharged into the circulation probably do not form viable metastases. Circulating cancer cells can be identified in patients who never develop clinical evidence of metastatic disease; presumably these cells die if they cannot implant or they are destroyed by the patient's immune system.

Cancer can spread as individual cells or cell clumps that migrate and implant. Whether spread occurs in groups or as individual cells there is still the problem of crossing the vascular endothelium (and basement membrane) to gain access to the tissue itself. Cancer cells probably implant themselves in distant tissues by exploiting, and subverting, the normal inflammatory response. By expressing inflammatory cytokines, cancer cells can deceive the endothelium of the host tissue into becoming activated and allowing cancer cells access to the extravascular space. Activated endothelium expresses receptors that bind to integrins and selectins on the surface of cells, allowing the cancer cells to move across the endothelial barrier.

Tumour-related inflammation

A malignancy can provoke an inflammatory response and the cytokines and other factors produced as a result of that response may act to promote and sustain malignant transformation. Growth factors, mutagenic reactive oxygen species, angiogenic factors and anti-apoptotic factors may all be produced as part of an inflammatory process and all may contribute to the progression of a cancer.

Evade detection/elimination

Although derived from normal cells ('self') cancer cells are, in terms of their genetic make-up, behaviour and characteristics, foreign ('not self'). As such, they ought to provoke an immune response and be eliminated. It is entirely possible that malignant transformation is a more frequent event than the emergence of clinical cancer. The possible role of the immune system in eliminating nascent cancers was proposed by Paul Ehrlich in 1909 and revisited by both Sir Frank McFarlane Burnet and Lewis Thomas in the late 1950s. Cancer cells, or at least those that give rise to clinical disease, appear to gain the ability to escape detection by the immune system. This may be through suppressing expression of tumour-associated antigens or it may be through actively co-opting one part of the immune system to help the tumour escape detection by other parts of the immune surveillance system. This hallmark has been exploited in recent years in the development of T-cell checkpoint inhibitors, which 'take the brakes' off the immune system to re-enable T-cell killing of cancer cells, e.g. in renal cell carcinoma, lung cancer and melanoma.

Loss of specialist cell function

Cancer cells are geared to excessive proliferation. They do not need to develop or retain those specialised functions that prior to malignant transformation were their physiological function. These cells can therefore afford to repress or permanently lose those genes that control such functions. The longer term disadvantage to this process is that cancer cells are vulnerable to external stressors, which may, in part, explain why some cancer treatments work.

Stephen Paget, 1855–1926, surgeon, The West London Hospital, London, UK. Paget's 'seed and soil' hypothesis is contained in his paper 'The distribution of secondary growths in cancer of the breast', published in the *Lancet* in 1889.

Paul Ehrlich, 1854–1915, Professor of Hygiene, the University of Berlin, and later Director of the Institute for Infectious Diseases, Berlin, Germany. In 1908, he shared the Nobel Prize in Physiology or Medicine with **Elie Metchnikoff**, 1845–1916, 'in recognition of his work on immunity'. Metchnikoff was Professor of Zoology at Odessa in Russia, and later worked at the Pasteur Institute in Paris, France.

Sir Frank McFarlane Burnet, 1899–1985, Australian virologist, Walter and Eliza Hall Institute, Melbourne, Australia. Burnett shared the 1960 Nobel Prize in Physiology or Medicine with **Sir Peter Brian Medawar**, 1915–1987, Jodrell Professor of Zoology, University College, London, UK, 'for their discovery of acquired immunological tolerance'.

Lewis Thomas, 1913–1993, American pathologist and immunologist, who became President of the Sloan Kettering Memorial Institute, New York, NY, USA.

Develop ability to change energy metabolism

Blood flow in malignant tumours is often sporadic and unreliable. As a result, cancer cells may have to spend prolonged periods in low-oxygen states (i.e. relative hypoxia). Compared with the corresponding normal cells, some cancer cells may be better able to survive in hypoxic conditions. This ability may enable tumours to grow and develop despite an impoverished blood supply. Cancer cells can alter their metabolism even when oxygen is abundant; they break down glucose but do not, as normal cells would do, send the resulting pyruvate to the mitochondria for conversion, in an oxygen-dependent process, to carbon dioxide. This is the phenomenon of aerobic glycolysis, or the Warburg effect, and leads to the production of lactate. In an act of symbiosis, lactate-producing cancer cells may provide lactate for adjacent cancer cells, which are then able to use it, via the citric acid cycle, for energy production. This cooperation is similar to that which occurs in skeletal muscle during exercise.

The growth of a cancer

If it is accepted that a cancer starts from a single transformed cell then it is possible, using straightforward arithmetic, to describe the progression from a single cell to a mass of cells large enough to kill the host. The division of a cell produces two daughter cells. The relationship $2n$ will describe the number of cells produced after n generations of division. There are between 10^{13} and 10^{14} cells in a typical human being. A tumour 10 mm in diameter will contain about 10^9 cells. Since $2^{30} = 10^9$ this implies that it would take 30 generations to reach the threshold of clinical detectability and, as $2^{45} = 3 \times 10^{13}$, it will take fewer than 15 subsequent generations to produce a tumour that, through sheer bulk alone, would be fatal. This is an oversimplification because cell loss is a feature of many cancers: for squamous cancers as many as 99% of the cells produced may be lost, mainly by exfoliation. It will, in the presence of cell loss, take many cellular divisions to produce a clinically evident tumour. The growth of a typical human tumour can be described by an exponential relationship, the doubling time of which increases exponentially – so-called Gompertzian growth (***Figure 12.2***).

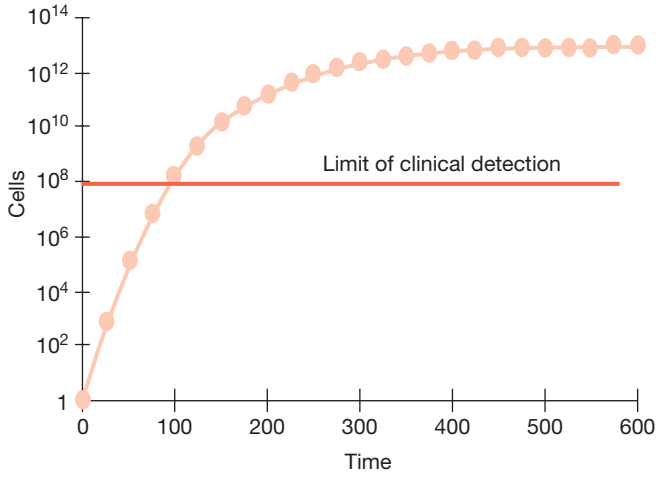


Figure 12.2 The Gompertzian curve describing the growth of a typical tumour. In its early stages, growth is exponential but, as the tumour grows, the growth rate slows. This decrease in growth rate probably arises because of difficulties with nutrition and oxygenation. The tumour cells are in competition: not only with the tissues of the host, but also with each other.

THE CAUSES OF CANCER

The interplay between nature and nurture

Both inheritance and environment are important determinants of cancer development. Neither influence is completely dominant. The balance between genes and the environment is context specific and not consistent. Two contrasting examples are breast and lung cancer. Although not all smokers develop lung cancer and lung cancer can occur in people who have never smoked, non-small cell lung cancer is much more common in smokers and is such a powerful risk factor that it accounts for approximately 80% of the disease. Conversely, germline *BRCA* gene mutations are highly penetrant and women with a *BRCA1* mutation can have a 60–90% lifetime risk of being diagnosed with breast cancer and 40–60% will develop ovarian cancer.

Knowledge about the causes of cancer can be used to design appropriate strategies for prevention or earlier diagnosis. As we find out more about the genes associated with cancer, genetic testing and counselling play increasingly important roles in the prevention of cancer. These considerations are incorporated into ***Table 12.1***, on the inherited cancer syndromes, and into ***Table 12.2***, on the environmental contribution to cancer.

THE MANAGEMENT OF CANCER

Management is more than treatment

The traditional approach to cancer concentrates on diagnosis and active treatment. This is a very limited view that, in terms of public health, may not have served society well. It implies a fatalistic attitude to the occurrence of cancer and an assumption that, once active treatment is complete, there is little more to be done. Prevention was forgotten and rehabilitation was ignored.

Otto Warburg, 1883–1970, chemist, Director of the Kaiser Wilhelm Institute for Cell Physiology, Berlin-Dahlem, Germany. Awarded the Nobel Prize in Physiology or Medicine in 1931 for 'his discovery of the nature and mode of action of the respiratory enzyme'.

Benjamin Gompertz, 1779–1865, an insurance actuary who described mathematically the relationship between life expectancy and age. The Gompertzian function provides an excellent fit to data points plotting tumour size against time.

TABLE 12.1 Examples of inherited syndromes associated with cancer.

Syndrome	Gene(s) implicated	Inheritance	Associated tumours and abnormalities	Strategies for prevention/ early diagnosis
Familial adenomatous polyposis (FAP)	*APC* gene	D	Colorectal cancer under the age of 25 Papillary carcinoma of the thyroid Cancer of the ampulla of Vater Hepatoblastomas Primary brain tumours (Turcot syndrome) Osteomas of the jaw CHRPE (congenital hypertrophy of the retinal pigment epithelium)	Prophylactic panproctocolectomy
Hereditary non-polyposis colorectal cancer (HNPCC1), Lynch syndrome 1	DNA mismatch repair genes (*MLH1; MSH2; MSH6*)	D	Colorectal cancer (typically in forties and fifties)	Surveillance colonoscopies/ polypectomies Non-steroidal anti-inflammatory drugs
HNPCC2	DNA mismatch repair genes *(MLH1; MSH2; MSH6)*	D	HNPCC associated with other cancers of the gastrointestinal or reproductive system	
Peutz–Jeghers syndrome	*STK11*	D	Bowel cancer; breast cancer; freckles round the mouth	Surveillance colonoscopy; mammography
Cowden[a] syndrome	*PTEN*	D	Multiple hamartomas of skin, breast and mucous membranes Breast cancer Neuroendocrine tumours Endometrial cancer Thyroid cancer	Active surveillance
Retinoblastoma	*RB*	D	Retinoblastoma Pinealoma Osteosarcoma	Surveillance of uninvolved eye
Multiple endocrine neoplasia (MEN) type 1	*Menin*	D	Parathyroid tumours Islet cell tumours Pituitary tumours	Awareness of associations and paying attention to relevant symptoms
MEN type 2A	*RET*	D	Medullary carcinoma of the thyroid Phaeochromocytoma Parathyroid tumours	Regular screening of blood pressure, serum calcitonin and urinary catecholamines Prophylactic thyroidectomy
MEN type 2B	*RET*	D	Medullary carcinoma of the thyroid Phaeochromocytoma Mucosal neuromas Ganglioneuromas of the gut	Regular screening of blood pressure, serum calcitonin and urinary catecholamines Prophylactic thyroidectomy
Li–Fraumeni	*P53*	D	Sarcomas Leukaemia Osteosarcomas Brain tumours Adrenocortical carcinomas	Very difficult, since pattern of tumours is so heterogeneous and varies between patients
Familial breast cancer	*BRCA1; BRCA2*	D	Breast cancer Ovarian cancer Papillary serous carcinoma of the peritoneum Prostate cancer	Screening mammography; pelvic ultrasound PSA (in males) Prophylactic mastectomy; prophylactic oophorectomy
Familial cutaneous malignant melanoma	*CDNK2A; CDK4*	D	Cutaneous malignant melanoma	Avoid exposure to sunlight, careful surveillance

Continued

Abraham Vater, 1684–1751, Professor of Anatomy and Botany, and later of Pathology and Therapeutics, Wittenberg, Germany.
Jacques Turcot, 1914–1977, surgeon, Hôtel-Dieu de Quebec hospital, Quebec, Canada.
Henry Thompson Lynch, 1928–2019, Chair of Preventative Medicine, Creighton University, Omaha, NE, USA.
John Law Augustine Peutz, 1886–1968, Chief Specialist for Internal Medicine, St John's Hospital, The Hague, The Netherlands.
Harold Joseph Jeghers, 1904–1990, Professor of Internal Medicine, The New Jersey College of Medicine and Dentistry, Jersey City, NJ, USA.
Frederick Pei Li, 1940–2015, Professor of Medicine, Harvard University Medical School, Boston, MA, USA.
Joseph F Fraumeni, b.1933, Director of Cancer Epidemiology and Genetics, The National Cancer Institute, Bethesda, MD, USA.

TABLE 12.1 Examples of inherited syndromes associated with cancer – *continued*.

Syndrome	Gene(s) implicated	Inheritance	Associated tumours and abnormalities	Strategies for prevention/ early diagnosis
Basal cell naevus syndrome (Gorlin)	*PTCH*	D	Basal cell carcinomas Medulloblastoma Bifid ribs	Careful surveillance, awareness of diagnosis (look for bifid ribs on x-ray)
von Hippel–Lindau disease	*VHL*	D	Clear cell renal cell carcinoma Phaeochromocytoma Haemangiomas of the cerebellum and retina	Urinary catecholamines
Neurofibromatosis type 1	*NF1*	D	Astrocytomas Primitive neuroectodermal tumours Optic gliomas Multiple neurofibromas	A difficult problem; maintain a high index of suspicion concerning any rapid changes in growth or character of any nodule
Neurofibromatosis type 2	*NF2*	D	Acoustic neuromas Spinal tumours Meningiomas Multiple neurofibromas	
Xeroderma pigmentosum	Deficient nucleotide excision repair (*XPA,B,C*)	R	Skin sensitive to sunlight. Early onset of cutaneous squamous or basal cell carcinomas	Avoidance of sun exposure Active surveillance and early treatment Retinoids for chemoprevention
Ataxia–telangiectasia	*AT*	R	Progressive cerebellar ataxia Leukaemia Lymphoma Breast cancer Melanoma Upper gastrointestinal tumours	Active surveillance
Bloom syndrome	*BLM helicase*	R	Sensitivity to ultraviolet light Leukaemia Lymphoma	Active surveillance

D, dominant; PSA, prostate-specific antigen; R, recessive.
[a]One of the few clinical syndromes named for the patient rather than the clinician. Rachel Cowden was, in 1963, the first patient described with the syndrome. She died from breast cancer at the age of 20.

TABLE 12.2 Environmental causes of cancer (and suggested measures for reducing their impact).

Environmental/behavioural factor		Associated tumours	Strategy for prevention/early diagnosis
Tobacco		Lung cancer Head and neck cancer Bladder cancer	Ban tobacco Ban smoking in public places Punitive taxes on tobacco
Alcohol		Head and neck cancer Oesophageal cancer Hepatocellular carcinoma	Avoid excess alcohol Surveillance of high-risk individuals
Ultraviolet exposure		Melanoma Non-melanoma skin cancer	Avoid excessive sun exposure, use high-factor sunscreen, avoid sunbeds
Ionising radiation		Leukaemia Breast cancer Lymphoma Thyroid cancer	Limit medical exposures to absolute minimum; safety precautions at nuclear facilities; monitor radiation workers
Viral infections	Human papillomavirus	Cervical cancer Penile cancer Head and neck cancer	Avoid unprotected sex Vaccination

Continued

Robert Gorlin, 1923–2006, Professor of Dentistry, The University of Minnesota, Minneapolis, MN, USA.
Eugen von Hippel, 1867–1939, Professor of Ophthalmology, Göttingen, Germany.
Arvid Lindau, 1892–1958, Professor of Pathology, Lund, Sweden.
David Bloom, 1892–1985, dermatologist at the Skin and Cancer Clinic, New York University, New York, NY, USA, described the syndrome in 1954.

Viral infections – *continued*	Human immunodeficiency virus		Kaposi's sarcoma Lymphomas Germ cell tumours Anal cancer	Avoid unprotected sex
	Hepatitis B		Hepatocellular carcinoma	Avoid contaminated injections/ infusions Vaccination
Other infections	Schistosomiasis		Bladder cancer	Treatment of infection
	Helicobacter pylori		Stomach cancer	Eradication therapy
Inhaled particles	Asbestos		Mesothelioma	Protect workers from inhaled dusts and fibres
	Wood dust		Paranasal sinus cancers	
Chemicals	Environmental pollutants/chemicals used in industry		Angiosarcoma (vinyl chloride) Bladder cancer (aniline dyes, vulcanisation of rubber) Lung, nasal cavity (nickel) Skin (arsenic) Lung (beryllium, cadmium, chromium) All sites (dioxins)	Protection of exposed workers; avoid chemical discharge and spillages
	Medical	Alkylating agents used in cytotoxic chemotherapy	Leukaemia Lymphoma Lung cancer	Avoid over-treatment; only combine drugs with ionising radiation when absolutely necessary
		Immunosuppressive treatment	Kaposi's sarcoma	As low a dose as possible, for as short a period as possible
		Tamoxifen	Endometrial cancer	Biopsy if patient on tamoxifen develops uterine bleeding
Fungal and plant toxins	Aflatoxins		Hepatocellular carcinoma	Appropriate food storage, screen for fungal contamination of foodstuffs
Obesity/lack of physical exercise			Breast Endometrium Kidney Colon Oesophagus	Maintain ideal body weight, regular exercise

A more comprehensive view considers the management of cancer as taking place along two axes: one is an axis of scale, from the individual to the world population; the other is an axis based on the development of the disease, from prevention through to rehabilitation or palliative care (***Figure 12.3***).

Prevention

There is much written on the evidence on the preventable causes of cancer. It is concluded that many cancers could be prevented if people ate sensibly, exercised more and avoided carcinogens such as cigarette smoke. Early identification of premalignant conditions may also prevent certain malignancies developing, e.g. Barrett's oesophagus as a premalignant condition in oesophageal cancer. This advice supplements the preventative measures outlined in ***Table 12.2***.

Screening

Screening involves the detection of disease in an asymptomatic population in order to improve outcomes by early diagnosis of cancer at a curable stage. It follows that a successful screening programme must achieve early diagnosis and that the disease in question has a better outcome when treated at an early stage. The criteria that must be fulfilled for the disease, screening test and the screening programme itself are given in ***Summary box 12.2***. Merely to prove that screening picks up disease at an early stage, and that the outcome is better for patients with screen-detected disease than for those who present with symptoms, is an insufficient criterion for the success of a screening programme. This is because of potential inherent biases of screening (lead time bias, selection bias and length bias), which make screen-detected disease appear to be associated with better outcomes than symptomatic disease.

Moritz Kaposi (originally Kohn), 1837–1902, Professor of Dermatology, Vienna, Austria, described pigmented sarcoma of the skin in 1872.
Norman Rupert Barrett, 1903–1979, surgeon, St Thomas' Hospital, London, UK.

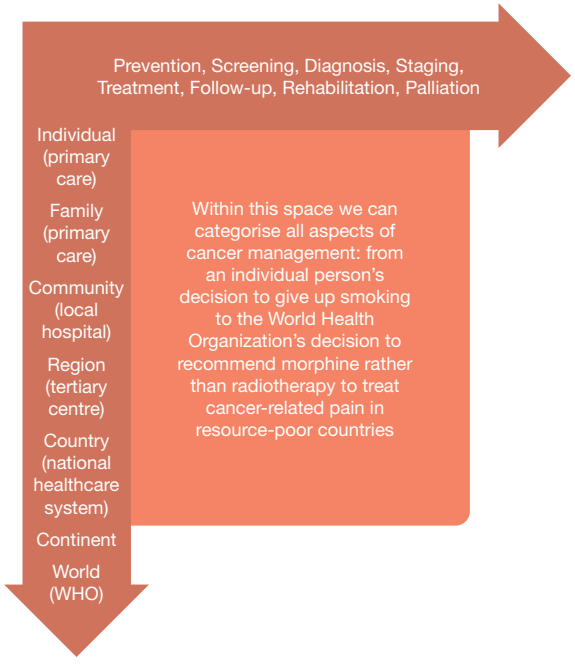


Figure 12.3 The management of cancer spans the natural history of the disease and all humankind, from the individual to the population of the world.

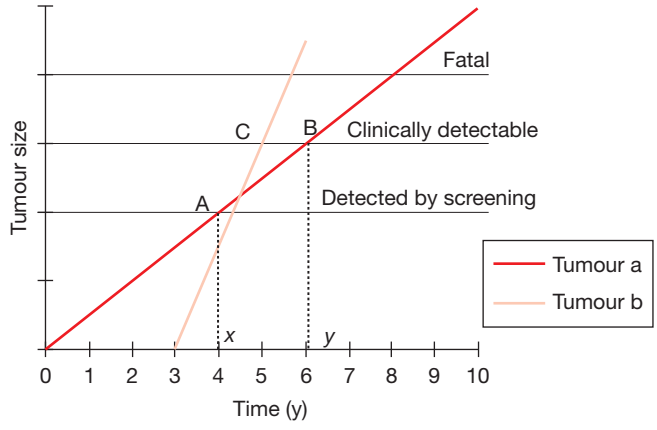


Figure 12.4 An illustration of lead time and length bias. ***Tumour a*** is a steadily growing tumour; its progress is uninfluenced by any treatment. Point A indicates the time at which the tumour would be diagnosed in a screening programme, and point B indicates the time at which the tumour would be diagnosed clinically, i.e. in the absence of any screening programme. If the date of diagnosis is used as the start time for measuring survival, then, in the absence of any effect from treatment, the screening programme will, artefactually, add to the survival time. The amount of 'increased' survival is equal to $y - x$ years, in this example just over 2 years. This artefactual inflation of survival time is referred to as lead time bias. ***Tumour b*** is a rapidly growing tumour; again its progress is uninfluenced by treatment. It grows so rapidly that, in the interval between two screening tests, it can cross both the threshold for detectability by screening and that of clinical detectability (at point C). It will continue to progress rapidly after diagnosis and the measured survival time will be short. This phenomenon, whereby those tumours that are 'missed' by the screening programme are associated with decreased survival, is called length time bias.

Summary box 12.2

Criteria for screening (based on Wilson–Junger criteria for a screening programme)

The disease:

- Recognisable early stage
- Treatment at early stage more effective than at later stage
- Sufficiently common to warrant screening

The test:

- Sensitive and specific
- Acceptable to the screened population
- Safe
- Inexpensive

The programme:

- Adequate diagnostic facilities for those with a positive test
- High-quality treatment for screen-detected disease to minimise morbidity and mortality
- Screening repeated at intervals if disease of insidious onset
- Benefit must outweigh physical and psychological harm

Lead time bias describes the phenomenon whereby early detection of a disease will always prolong survival from the time of diagnosis when compared with disease picked up at a later stage in its development whether or not detection in the screening process has altered the progression of the cancer (*Figure 12.4*). Selection bias describes the finding that individuals who accept an invitation for screening are, in general, healthier than those who do not. It follows that individuals with screen-detected disease will tend to live longer, independently of the condition for which screening is being performed. Length bias occurs because small, slow-growing tumours are likely to be picked up by screening whereas larger, fast-growing tumours are likely to arise and produce symptoms in between screening rounds. Screen-detected tumours will therefore tend to be less aggressive than symptomatic tumours. Because of these biases it is essential to carry out population-based randomised controlled trials and to compare the mortality rate in a whole population offered screening (including those who refuse to be screened and those who develop cancer after a negative test) with the mortality rate in a population that has not been offered screening. This research design has been applied to both breast cancer and colorectal cancer: in both cases there was a reduction in disease-specific mortality. However, in general, clinical trials of screening with the gold-standard endpoint of overall survival have not been undertaken because of the very large number of participants required with long study follow-up periods.

Cancer screening remains a controversial topic with advocates on both sides of the argument. Targeted, risk-based screening approaches, such as computed tomography (CT) scan-based screening of smokers and ex-smokers for lung cancer, are being evaluated as methods of developing more conclusive screening programmes.

James Maxwell Glover Wilson, 1913-2006, Principal Medical Officer at the Ministry of Health in London, England.
Gunnar Jungner, 1914–1982, Chief Clinical Chemistry, Sahlgrenska Hospital, Gothenburg, Sweden.

Diagnosis and classification

Accurate diagnosis is the key to successful management of cancer. Precise diagnosis is crucial to the choice of correct therapy; the wrong operation, no matter how well performed, is useless. An unequivocal diagnosis is also the key to an accurate prognosis. Only rarely can a diagnosis of cancer be confidently made in the absence of tissue for pathological or cytological examination. Cancer is a disease of cells and, for accurate diagnosis, the abnormal cells need to be obtained and visualised by a histopathologist. Different tumours are classified in different ways: most squamous epithelial tumours are classed as well (G1), moderate (G2) or poorly (G3) differentiated. Adenocarcinomas are also often classified as G1, G2 or G3 but prostate cancer is an exception, with widespread use of the Gleason system. The Gleason system grades prostate cancer according to the degree of differentiation of the two most prevalent architectural patterns. The final score is the sum of the two grades and can vary from 6 (3 + 3) to 10 (5 + 5) with the higher scores indicating poorer prognosis.

The ongoing development of molecular classifiers in many cancer types is beginning to profoundly alter our approach to treatment of these malignancies based on genetic mutations and other molecular features identified in individual patients, i.e. in melanoma where patients with *BRAF* gene mutations can be successfully treated with the *BRAF* inhibitors. Molecular characterisation of malignancies and identification of their vulnerabilities have already become standard of care in many cancers (i.e. breast, colorectal and lung cancer) and this list is likely to expand.

Cancer staging

It is not sufficient simply to know what and where a cancer is; its extent must also be known. If it is localised, then locoregional treatments such as surgery and radiation therapy may be curative. If the disease is widespread then, although such local interventions may contribute to cure, they will be insufficient and systemic treatment, for example with drugs or hormones, will also be required.

Staging is the process whereby the extent of disease is mapped out. Staging used to be a fairly crude process based on clinical examination, chest x-ray and occasionally ultrasound; it is now a sophisticated process, reliant on advanced imaging techniques such as CT, magnetic resonance imaging (MRI) and positron emission tomography (PET) scans. These technological advances bring with them the implication that patients staged as having localised disease today are not comparable to patients described in 1985 as having localised disease. Many of these latter patients would, had they been imaged using the technology of today, have had occult metastatic disease detected.

The Union for International Cancer Control (UICC) is responsible for the TNM (tumour, nodes, metastases) staging system for cancer. This system is compatible with, and relates to, the American Joint Committee on Cancer (AJCC) system

TABLE 12.3 Staging of colorectal cancer.

TNM		
TX,	Primary tumour cannot be assessed	
T0,	No evidence of primary tumour	
Tis,	Carcinoma *in situ* or intramucosal carcinoma	
T1,	Tumour invades submucosa	
T2,	Tumour invades muscularis propria	
T3,	The tumour has grown through the muscularis propria and into the subserosa, which is a thin layer of connective tissue beneath the outer layer of some parts of the large intestine, or it has grown into tissues surrounding the colon or rectum	
T4,	Tumour directly invades beyond bowel	a, The tumour has grown into the surface of the visceral peritoneum
		b, The tumour has grown into or has attached to other organs or structures
NX,	Regional lymph nodes cannot be assessed	
N0,	No metastases in regional nodes	
N1a,	Metastases in 1 regional lymph node	
N1b,	Metastases in 2 or 3 regional lymph nodes	
N1c,	There are nodules made up of tumour cells found in the structures near the colon that do not appear to be lymph nodes	
N2a,	Metastases in 4–6 regional lymph nodes	
N2a,	Metastases in ≥7 regional lymph nodes	
MX,	Not possible to assess the presence of distant metastases	
M0,	No distant metastases	
M1a,	The cancer has spread to 1 other part of the body beyond the colon or rectum	
M1b,	The cancer has spread to more than 1 part of the body other than the colon or rectum	
M1c,	The cancer has spread to the peritoneal surface	

Donald F Gleason, 1920–2008, pathologist, The University of Minnesota, Minneapolis, MN, USA.

for stage grouping of cancer. As an example, the TNM clinicopathological staging system for colorectal cancer is shown in *Table 12.3*.

The purpose of staging is twofold: to estimate prognosis and to help select appropriate treatment options. Anatomical staging provides important information as to the surgical resectability of disease and the risk of its recurrence. However, the risk assessment for most cancers is suboptimal and provides a wide confidence interval. Better staging techniques are entering care, such as the determination of circulating tumour DNA (ctDNA) in the blood of postoperative patients. Patients who demonstrate ctDNA postoperatively have a substantially higher risk of relapse than patients who do not. This information may be useful in the future to select patients who should be offered further treatment to eliminate residual tumour cells and, of equal importance, to identify patients with a low chance of relapse who can avoid further treatment.

Therapeutic decision making and the multidisciplinary team

As the management of cancer becomes more complex, it becomes impossible for an individual clinician to have the competence that is necessary to manage all patients presenting with a particular type of tumour. The formation of multidisciplinary teams represents an attempt to make certain that each and every patient with a particular type of cancer is managed appropriately. Teams should not only be multidisciplinary – they should also be multiprofessional. The advantages and disadvantages of multidisciplinary teams are summarised in *Table 12.4*.

The multidisciplinary team needs to answer three basic questions for every patient:

- What is the patient's diagnosis, stage and molecular characteristics of disease?
- What is the goal of treatment for the patient? In simple terms, this can be divided into cure, prolongation of life and palliation of symptoms.
- What treatment options are there to achieve these aims with the fewest possible side effects? Options may include surgery, radiotherapy, anti-cancer systemic (drug) treatment, symptom control measures, a combination of these options and, in some cases, observation.

It is important to realise that the multidisciplinary team makes recommendations, rather than definite treatment decisions. Several other factors must be considered, which can only be done by a clinician making a direct assessment of the patient's health and wishes. The clinician must ascertain the patient's fitness and hence their ability to tolerate treatment. This will be heavily influenced by comorbidities. Patients vary greatly in their willingness to tolerate treatment for a given benefit and particularly how they value quality of life compared with length of life or chance of cure.

There are often several possible treatment plans and the clinical team must take the time to explain the options carefully to the patient and the patient's supporters. In many cancer centres, there will be both standard-of-care and research options available for patients. Explanations should include what is involved in the treatment, what benefits may result and the chance of the patient receiving those benefits. The clinical team must also explain the possible downsides of treatment and the likelihood of experiencing them. Patients are often faced with a large amount of complex and difficult information at a time when they are extremely vulnerable. The clinical team must support patients to reach a decision and this may take time and repeated explanation.

Principles of cancer surgery

For most solid tumours, surgery remains the definitive treatment and the only realistic hope of cure. However, surgery has many roles in cancer treatment from diagnosis, prevention, removal of primary disease, removal of metastatic disease, reconstruction through to palliation of symptoms.

Role in diagnosis and staging

In most, but not all, patients the diagnosis of cancer has been confirmed by biopsy before definitive surgery is carried out; however, occasionally a surgical procedure is required to make the diagnosis, e.g. in renal cell cancer where not all patients with a renal mass will undergo a biopsy, prior to definitive surgery to remove and diagnose the tumour as either malignant or benign. Laparoscopic surgery is used as part of the staging of intra-abdominal malignancy, particularly oesophageal and gastric cancer. By this means it is often possible to diagnose widespread peritoneal disease and small liver metastases that may have been missed on cross-sectional imaging. Laparoscopic

TABLE 12.4 The advantages and disadvantages of the multidisciplinary team.

Advantages	Disadvantages
Open debate concerning management of complex patients with many specialists	Less confident and less articulate members of the team may not be able to express their views, even though their views may be extremely important
Decision making is open, transparent and explicit	May become a rubber-stamping exercise in which the class solutions implied by guidelines are applied to disparate individuals
Team members educate each other	Decisions are made in the absence of patients and their carers
A useful educational experience for trainees and students	Clinicians are able to avoid having to take responsibility for their decisions and their actions – 'corporate responsibility'
Performance can be monitored by managers	Time-consuming and resource intensive: takes multiple busy clinicians away from clinical practice for hours at a time

Summary box 12.3

Potential members of the cancer multidisciplinary team

- Site-specialist surgeon
- Surgical oncologist
- Plastic and reconstructive surgeon
- Clinical oncologist/radiotherapist
- Medical oncologist
- Diagnostic radiologist
- Interventional radiologist
- Palliative care physician
- Pathologist
- Speech therapist
- Physiotherapist
- Prosthetist
- Clinical nurse specialist (rehabilitation, supportive care)
- Clinical trial team representative
- Palliative care nurse
- Social worker/counsellor
- Medical secretary/administrator
- Multidisciplinary team coordinator

ultrasound is a particularly useful adjunct for the diagnosis of intrahepatic metastases. Other examples of when surgery is central to the diagnosis of cancer include orchidectomy, in a patient suspected of testicular cancer; lymph node biopsy in a patient with lymphoma; and sentinel node biopsy in melanoma and breast cancer.

Removal of primary disease

Radical surgery for cancer involves removal of the primary tumour and as much of the surrounding tissue and lymph node drainage as possible in order not only to ensure local control but also to prevent spread of tumour through the lymphatics. Although the principle of local control is still extremely important, it is now recognised that ultra-radical surgery probably has little effect on the development of metastatic disease, as evidenced by the randomised trials of radical versus simple mastectomy for breast cancer. It is important, however, to appreciate that high-quality, meticulous surgery taking care not to disrupt the primary tumour at the time of excision is of the utmost importance in obtaining a cure in localised disease and in preventing local recurrence.

Removal of metastatic disease

Under certain circumstances surgery for metastatic disease may be appropriate. This is particularly true for liver metastases arising from colorectal cancer where successful resection of all detectable disease can lead to long-term survival in about one-third of patients. With multiple liver metastases, it may still be possible to take a surgical approach by using *in situ* ablation with cryotherapy or radiofrequency energy. Another situation in which surgery may be curative in metastatic disease is that of pulmonary resection for isolated lung metastases, particularly from renal cell carcinoma.

Palliation

In many cases surgery is not appropriate for cure but may be valuable for palliation. A good example of this is the patient with a symptomatic primary tumour who also has distant metastases, e.g. a patient with a large, symptomatic renal cell cancer but diffuse metastatic disease. In this case, removal of the primary will increase the patient's quality of life but will have little effect on the ultimate outcome.

Principles underlying the non-surgical treatment of cancer

Medical and clinical (radiation) oncology are rapidly changing fields. Both understanding of cancer and the technology available are expanding at a rapid pace. Nevertheless, there are basic principles that remain constant. These are to ascertain as precisely as possible the diagnosis, stage of disease and molecular characteristics of the tumour and, from this information, assess via the multidisciplinary team the management options for the patient. In general terms, where a local treatment, i.e. surgery or ablation, is as effective as a systemic treatment, the local treatment will be preferred. The range of options will be specific to each tumour type and is constantly evolving. This is most true of the systemic therapy options available to patients.

The purposes of oncology treatment can be conveniently divided into:

- **curative primary**: the effect of non-surgical treatment alone is highly effective, e.g. head and neck cancers (*Table 12.5*);
- **neoadjuvant**: where non-surgical treatment prior to surgery can substantially reduce the morbidity of treatment and increase its chances of success, e.g. radiotherapy to downstage rectal carcinoma prior to surgery;
- **adjuvant**: where non-surgical treatment after surgery can increase the chance of cure, e.g. radiotherapy and chemotherapy after breast cancer;
- **life-prolonging/palliative**: these terms are better separated, especially in the case of life-prolonging treatments in a non-curative setting that may add years to life, e.g. use of olaparib in *BRCA*-mutated ovarian cancer.

Systemic therapies

In recent decades, drug development has evolved from screening large libraries of chemicals for their ability to interfere with cellular processes. Increased understanding of the molecular pathophysiology of cancers has identified key molecules essential to tumour function. Many of these molecules can be inhibited and are termed 'druggable targets'. Three-dimensional characterisation of these targets and synthesis of molecules to inhibit them has produced a large number of highly effective targeted therapies. The pace of change is sufficiently rapid that it is now only possible for an individual clinician to keep abreast of developments in a limited number of cancers.

The following principles are key in decision making over systemic therapy administration to individual patients:

TABLE 12.5 Examples of malignancies that may be cured without the need for surgical excision.

Malignancy	Potentially curative treatment
Leukaemia	Chemotherapy (+/– radiotherapy)
Lymphoma	Chemotherapy (+/– radiotherapy)
Small cell lung cancer	Chemotherapy (+/– radiotherapy)
Tumours of childhood (rhabdomyosarcoma, Wilms' tumour)	Chemotherapy (+/– radiotherapy)
Early laryngeal cancer	Radiotherapy
Advanced head and neck cancer	Chemoradiation (synchronous chemotherapy and radiotherapy)
Oesophageal cancer	Chemoradiation (synchronous chemotherapy and radiotherapy)
Squamous cell cancer of the anus	Chemoradiation (synchronous chemotherapy and radiotherapy)
Advanced cancer of the cervix	Radiotherapy (+/– chemotherapy)
Medulloblastoma	Radiotherapy (+/– chemotherapy)
Skin tumours (BCC, SCC)	Radiotherapy

BCC, basal cell carcinoma; SCC, squamous cell carcinoma.

- Assess the fitness and willingness of the patient to tolerate each of these options.
  Most cancers occur in middle-aged or older patients. The older a patient is, the more likely they are to have comorbidities or be frail. It is important to note that middle-aged patients may be unfit or have serious comorbidities and that older patients may actually be very fit. Organ dysfunction and frailty may substantially increase the risks of treatment and reduce its chance of success. This must be assessed on an individual basis.
- Support the patient to understand the options and choose the most appropriate management approach.
  Discussions about prognosis and treatment options are complex, difficult and any decisions made are often irrevocable. They also take place when the patient and their family are at their most anxious and vulnerable. It is often necessary to provide information in digestible amounts and to do so repeatedly. There are few oncological situations where an immediate decision needs to be taken by the patient.
  The difficulties of drawing a balance between risk and benefit are illustrated by considering adjuvant therapy for a solid malignancy. In the absence of a good test to ascertain individual risk of relapse, it is necessary to treat large numbers of patients to benefit a small proportion, often around 10% (*Figure 12.5*). Although 10% is a small proportion, it represents a small chance of a major benefit, namely cure. If there were a test that could accurately determine patients with no residual disease after resection, then a lower proportion of patients would need to have adjuvant therapy, each of whom would have a higher chance of benefit (*Figures 12.5 and 12.6*).
- Frequently reassess the balance of risk and benefit during treatment.
  For patients receiving repeated cycles of treatment, there are three key questions to consider before proceeding with the next cycle of treatment:

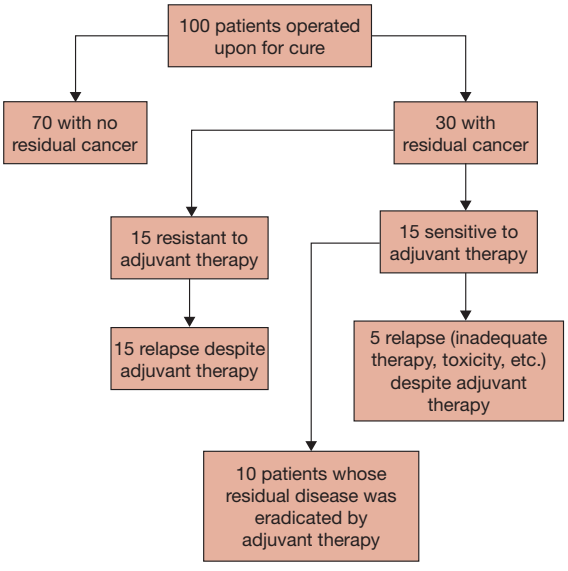
  1. **Is the treatment working**? Where there is measurable disease, this is usually assessed by radiological restaging at intervals of 6–12 weeks. The test for whether the treatment is working will depend on the goals of treatment: if the goal is stabilisation of disease then it may be sufficient that the tumour is not growing; if the goal is elimination of disease then progressive shrinkage of the radiological abnormalities will be necessary.
  2. **Is the patient tolerating treatment?** This can only be discovered by asking the patient what side effects they are experiencing. Clinicians should be used to detecting and managing the side effects of the drugs they are prescribing. In addition to open questions, the clinician should enquire specifically about common or dangerous side effects. If the patient is not tolerating treatment, then it may be necessary to delay the next cycle or to reduce the dose, even though this may be at the expense of reducing the effectiveness of treatment.
  3. **Is it safe to give the next cycle of treatment?** To answer this question, the clinician will have to assess what side effects the patient is experiencing and also ensure that laboratory measures are within acceptable limits. The necessary laboratory tests will commonly include full blood count, urea and electrolytes and liver function tests as well as other tests determined by which treatments are being used.

The range of non-surgical anti-cancer treatments

Radiotherapy

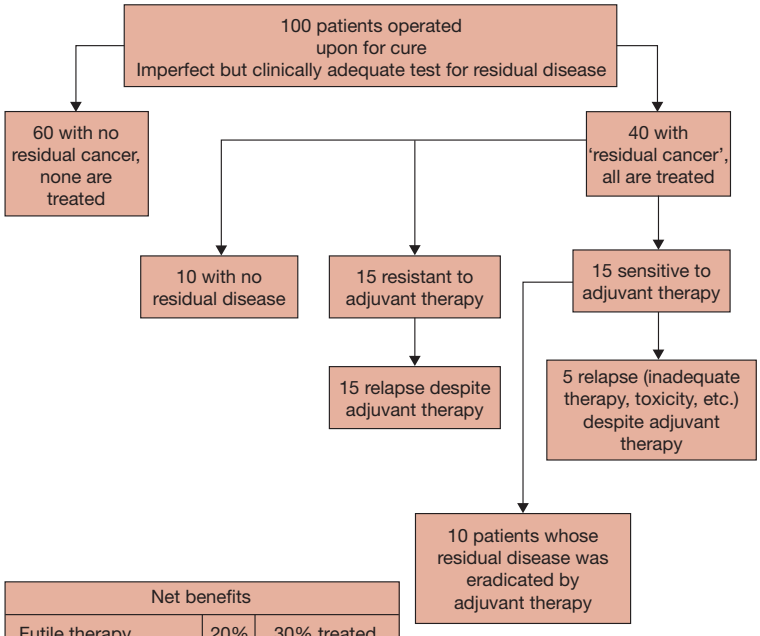
Half of all patients will receive radiotherapy during their cancer journey. The mechanism of action of radiotherapy is that ionising radiation causes single- and double-stranded DNA breaks. Cells most sensitive to radiotherapy are in mitosis or late G2 in the cell cycle. Damage may be caused by the direct effect of the ionising radiation on the DNA. However, the predominant therapeutic effect is indirect, and is achieved by ionising radiation causing the creation of oxygen free radicals, which then damage the DNA. The fact that the indirect damage is predominant explains why hypoxic cells are relatively resistant to radiotherapy. In practice, adequate

Max Wilms, 1867–1918, Professor of Surgery, Heidelberg, Germany.



Net benefits		
Futile therapy	20%	90% treated inappropriately
Unnecessary therapy	70%	
Beneficial therapy	10%	10% treated appropriately
No therapy	0%	

Figure 12.5 The concept of adjuvant chemotherapy.



Net benefits		
Futile therapy	20%	30% treated inappropriately
Unnecessary therapy	10%	
Beneficial therapy	10%	70% treated appropriately
No therapy	60%	

Figure 12.6 The concept of adjuvant chemotherapy and testing for minimal residual disease.

oxygenation of tissues is assisted by simple measures such as correcting anaemia prior to radiotherapy.

Cancer types differ greatly in their sensitivity to radiotherapy. Similarly, higher doses of radiotherapy are needed to control bulky disease (~70 Gy) than to eliminate residual tumour cells (~50 Gy) or to palliate symptoms (~30 Gy).

Radiotherapy is mainly used as a localised treatment (*Figure 12.7*). The target area is defined by imaging, which will include anatomical imaging, such as CT and MRI, to identify the tumour mass, neighbouring structures and landmarks for radiotherapy planning. Sometimes functional imaging, such as PET, is used to further define structures that should be included in the radiation field such as lymph nodes containing small volumes of tumour, whose metabolic signature is detected by PET. The planning and technical set-up of radiotherapy is a highly specialised subject. Planning can be extremely difficult and time-consuming, e.g. in the treatment of head and neck cancer, where the treatment intent is curative but there are multiple vital and highly sensitive structures close to the target area.

In general terms, the radiotherapy team will aim to provide a uniform therapeutic radiation dose to the target area, while sparing adjacent structures from significant damage. The most common form of treatment is external beam radiotherapy in which x-rays are aimed at the target volume from an external source. Where necessary, modern radiotherapy can employ sophisticated computer programs and algorithms to sculpt the shape of the treatment volume. It is now possible to irradiate irregularly shaped target volumes using techniques such as intensity-modulated radiation therapy (IMRT) and image-guided radiation therapy (IGRT). The ability to give high fractional doses in this way has led to the development of stereotactic ablative radiotherapy (SAbR), in which a small number of fractions can be used to treat primary tumours, or isolated metastases, to curative dose levels.

Other ways of targeting the tumour volume include brachytherapy, in which the radiation source is brought very close to the tumour by insertion of radioactive seeds, and radionuclide therapy, in which a radioactive source is concentrated at the tumour site.

One of the main problems in determining the optimal schedule of radiation is that there is a dissociation between the acute effects on normal tissues and the late damage. The acute reaction is not a reliable guide to the adverse consequences of treatment in the longer term. Since the late effects following irradiation can take over 20 years to develop, this poses an obvious difficulty: if a radiation schedule is changed it will be known within 2 or 3 years whether or not the new schedule has improved tumour control; it may, however, be two decades before it is known, with any degree of certainty, whether or not the new technique is safe.

Anti-cancer drugs

The classes of anti-cancer drugs, their modes of action and clinical indications are summarised in *Table 12.6*.

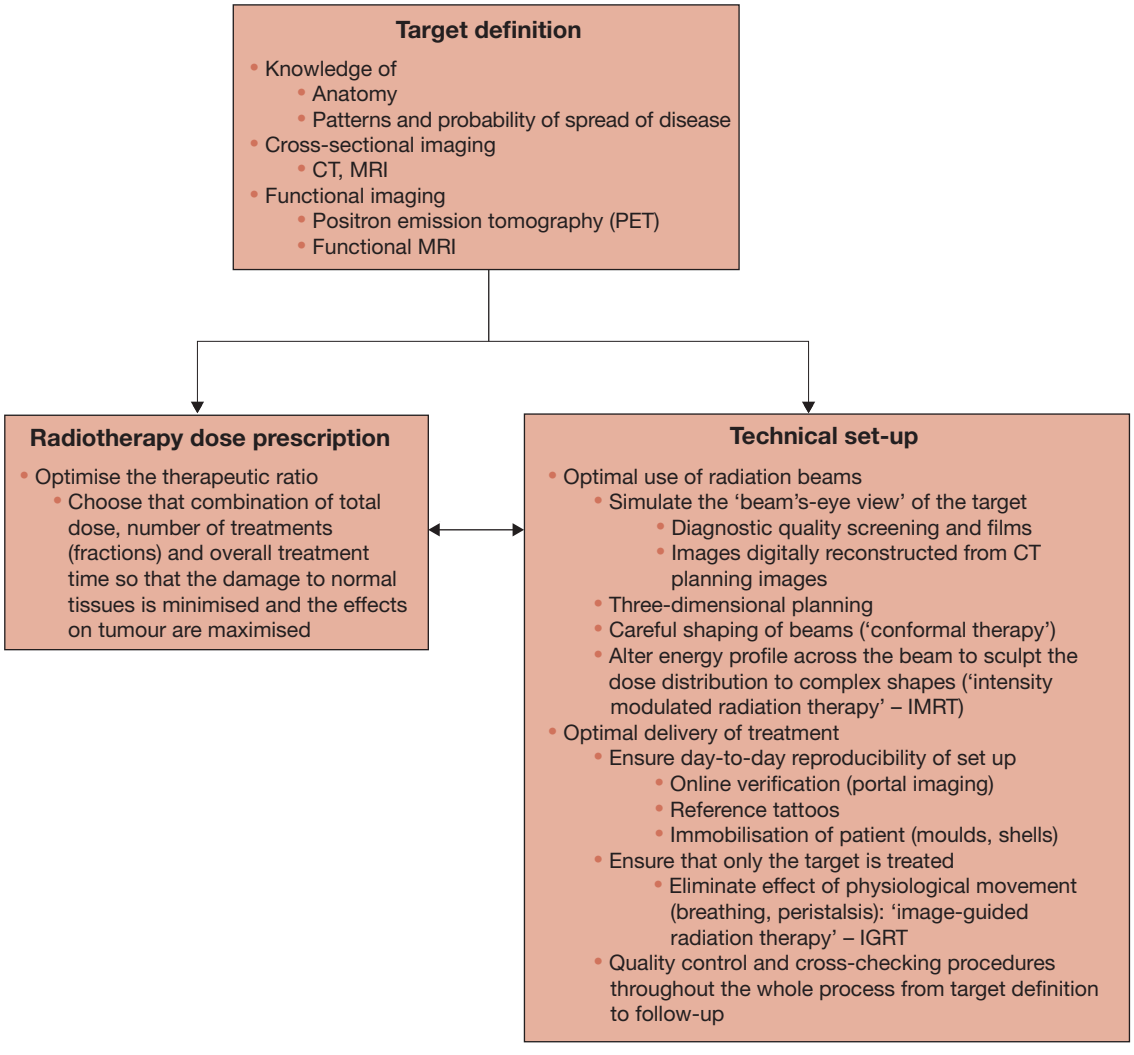


Figure 12.7 The processes involved in clinical radiotherapy. CT, computed tomography; MRI, magnetic resonance imaging.

TABLE 12.6 Examples of anti-cancer drugs currently in use.

Class	Examples	Putative mode of action	Tumour types that may be sensitive to drug
Drugs that interfere with mitosis	Vincristine Vinblastine	Interfere with formation of microtubules: 'spindle poisons'	Lymphomas Leukaemias Brain tumours Sarcomas
	Taxanes: paclitaxel docetaxel cabazitaxel	Stabilise microtubules	Breast cancer Non-small cell lung cancer Ovarian cancer Prostate cancer Head and neck cancer
Drugs that interfere with DNA synthesis (antimetabolites)	5-Fluorouracil Capecitabine	Inhibition of thymidylate synthase, false substrate for both DNA and RNA synthesis	Breast cancer GI cancer
	Methotrexate	Inhibition of dihydrofolate reductase	Breast cancer Bladder cancer Lymphomas Cervical cancer
	6-Mercaptopurine 6-Thioguanine	Inhibit *de novo* purine synthesis	Leukaemias

Continued

TABLE 12.6 Examples of anti-cancer drugs currently in use – *continued*.

Class	Examples	Putative mode of action	Tumour types that may be sensitive to drug
Drugs that interfere with DNA synthesis (antimetabolites) – *continued*	Cytosine arabinoside	False substrate in DNA synthesis	Leukaemias Lymphomas
	Gemcitabine	Inhibits ribonucleotide reductase	Non-small lung cancer Pancreatic cancer
Drugs that directly damage DNA or interfere with its function	Mitomycin C	DNA cross-linking, preferentially active at sites of low oxygen tension (a bioreductive drug)	Anal cancer Bladder cancer Gastric cancer Head and neck cancer Rectal cancer
	Cisplatin Carboplatin	Form adducts between DNA strands and interferes with replication	Germ cell tumours Ovarian cancer Non-small cell lung cancer Head and neck cancer Oesophageal cancer
	Oxaliplatin	Forms adducts between DNA strands and interferes with replication	Colorectal cancer
	Doxorubicin	Intercalates between DNA strands and interferes with replication	Breast cancer Lymphomas Sarcomas Kaposi's sarcoma
	Cyclophosphamide	A prodrug converted via hepatic cytochrome p450 to phosphoramide mustard. Causes DNA cross-links	Breast cancer Lymphomas Sarcomas
	Ifosfamide	Related to cyclophosphamide, causes DNA cross-links	Small cell lung cancer Sarcomas
	Bleomycin	DNA strand breakage via formation of metal complex	Germ cell tumours Lymphomas
	Irinotecan	Inhibits topoisomerase 1, prevents DNA from unwinding and repairing during replication	Colorectal cancer
	Etoposide	Inhibits topoisomerase 2, prevents DNA from unwinding and repairing during replication	Small cell lung cancer Germ cell tumours Lymphomas
	Dacarbazine	A nitrosourea that requires activation by hepatic cytochrome p450. Methylates guanine residues in DNA	Brain tumours Sarcoma
	Temozolomide	A nitrosourea but, unlike dacarbazine, does not require activation by hepatic cytochrome p450. Methylates guanine residues in DNA	Glioblastoma multiforme
	Actinomycin D	Intercalation between DNA strands, DNA strand breaks	Rhabdomyosarcoma Wilms' tumour
DNA damage response inhibitors	Olaparib Niraparib Rucaparib	Inhibit DNA repair by PARP, especially in *BRCA*-mutated cancers	Ovarian cancer Breast cancer Prostate cancer
Hormones	Tamoxifen	Blocks oestrogen receptors	Breast cancer
	Anastrozole Letrozole Exemestane	Aromatase inhibitors that block postmenopausal (non-ovarian) oestrogen production	Breast cancer
	Leuprolide Goserelin Buserelin	Analogues of gonadotropin-releasing hormone, continued use produces downregulation of the anterior pituitary with consequent fall in testosterone levels	Prostate cancer
	Cabergoline	Blocks prolactin release, a long-acting dopamine agonist	Prolactin-secreting pituitary tumours

Continued

TABLE 12.6 Examples of anti-cancer drugs currently in use – *continued*.

Class	Examples	Putative mode of action	Tumour types that may be sensitive to drug
Hormones – *continued*	Bromocriptine	Dopamine agonist, blocks stimulation of anterior pituitary	Pituitary tumours
	Cyproterone acetate Flutamide Nilutamide Bicalutamide	Block the effect of androgens	Prostate cancer
	Abiraterone Cyproterone acetate	Block testosterone production	Prostate cancer
Inhibitors of receptor tyrosine kinases	Osimertinib Erlotinib Afatinib Gefitinib	Inhibit EGFR tyrosine kinase	Non-small cell lung cancer
	Imatinib	Blocks ability of mutant BCR-ABL fusion protein to bind ATP	Chronic myeloid leukaemia
	Imatinib	Inhibition of mutant c-KIT	GISTs
	Erlotinib	Inhibits EGFR tyrosine kinase	Non-small cell lung cancer Pancreatic cancer
	Sunitinib Regorafenib Lenvatinib	Promiscuous tyrosine kinase inhibitors (PDGFR, VEGFR, KIT, FLT)	Renal cancer GIST refractory to Imatinib Colorectal cancer Thyroid cancer
	Lapatinib	Inhibits tyrosine kinases associated with EGFR and HER2	Breast cancer
	Axitinib	Inhibits tyrosine kinase associated with VEGFR	Renal cancer
Cyclin-dependent kinase inhibitors	Palbociclib	Inhibits growth signal	Breast cancer
Protease inhibitors	Bortezomib	Interferes with proteasomal degradation of regulatory proteins, in particular stops NF-κB from preventing apoptosis	Multiple myeloma
Differentiating agents	All-*trans*-retinoic acid	Induces terminal differentiation	Acute promyelocytic leukaemia
Farnesyl transferase inhibitors	Lonafarnib	Inhibition of farnesyl transferase and consequent inactivation of *ras*-dependent signal transduction	Leukaemia
	Tipifarnib	Inhibition of farnesyl transferase and consequent inactivation of *ras*-dependent signal transduction	Acute leukaemia Myelodysplastic syndrome
Antibodies directed to cell surface antigens	Trastuzumab	Antibody directed against HER2 receptor	Breast cancer
	Cetuximab	Antibody directed against EGFR	Colorectal cancer Head and neck cancer
	Bevacizumab	Antibody directed against VEGFR	Colorectal cancer
	Rituximab	Antibody against CD20 antigen	Lymphomas
	Alemtuzumab	Antibody against CD52 antigen	Lymphomas
Antibody drug conjugates	Trastuzumab deruxtecan	Targets chemotherapy to *Her2* expressing tumour	Breast cancer
	Sacituzumab govitecan	Targets chemotherapy to *Trop2* expressing tumour	Breast cancer
	Enfortumab vedotin	Targets chemotherapy to *Nectin4* expressing tumour	Urothelial cancer
Inducers of apoptosis	Arsenic trioxide	Induces apoptosis by caspase inhibition Inhibition of nitric oxide	Acute promyelocytic leukaemia
	Venetoclax	BH3 mimetic	CML AML Lymphoma

Continued

TABLE 12.6 Examples of anti-cancer drugs currently in use – *continued*.

Class	Examples	Putative mode of action	Tumour types that may be sensitive to drug
Immunological mediators	Ipilimumab	Blocks CTLA-4 and thus releases the brakes on the activation of T cells	Melanoma Lung cancer Renal cancer
	Pembrolizumab Nivolumab Atezolizumab Durvalumab	Block the PD-1 signalling on T lymphocytes and thereby prevents the inhibition of T-cell activation	Melanoma Lung cancer Renal cancer Bladder cancer
	Interferon alpha-2b	Activates macrophages, increases the cytotoxicity of T lymphocytes, inhibits cell division (and viral replication)	Hairy-cell leukaemia
	Thalidomide	Anti-inflammatory, stimulates T cells, antiangiogenic	Myeloma
HDAC inhibitors	Panobinostat Vorinostat	Acetylation of histones is associated with increased transcription of genes; inhibiting deacetylation can decrease expression of mutated or dysregulated genes	Cutaneous T-cell lymphoma
	Entinostat	Acetylation of histones is associated with increased transcription of genes; inhibiting deacetylation can decrease expression of mutated or dysregulated genes	Melanoma
PI3K inhibitors	Idelalisib	Inhibits signalling via the PI3K/AKT/mTOR pathway and thereby switch off stimulus to cellular proliferation	B-cell lymphomas
mTOR inhibitors	Temsirolimus Everolimus	Inhibit mTOR, a key component in the PI3K/AKT/mTOR pathway	Renal cancer Neuroendocrine tumours
MEK inhibitors	Trametinib Selumetinib	Inhibit the MAPK pathway	Melanoma Neurofibroma
RAF inhibitors	Dabrafenib Vemurafenib	Inhibit the MAPK pathway	Melanoma

AML, acute myeloid leukaemia; ATP, adenosine triphosphate; CML, chronic myeloid leukaemia; CTLA-4, cytotoxic T-lymphocyte-associated protein 4; EGFR, epidermal growth factor receptor; FLT3, FMS-like tyrosine kinase; GI, gastrointestinal; GIST, gastrointestinal stromal tumour; HDAC, histone deacetylase; HER2, human epidermal growth factor receptor 2; MAPK, mitogen-activated protein kinase; mTOR, mammalian target of rapamycin; NF, nuclear factor; PARP, poly-ADP ribose polymerase; PD-1, programmed cell death protein 1; PDGFR, platelet-derived growth factor receptor; PI3K, phosphoinositide 3-kinase; RAF, rapidly accelerated fibrosarcoma; TKI, tyrosine kinase inhibitor; VEGFR, vascular endothelial growth factor receptor.

Cytotoxic chemotherapy

Selective toxicity is the fundamental principle underlying cytotoxic chemotherapy. Tumour types vary greatly in their sensitivity to cytotoxic chemotherapy and their vulnerability to drugs with specific mechanisms of action. Treatments are often given for a limited number of treatment days during a cycle of treatment lasting typically 3–4 weeks. This allows the tumour to receive an effective dose while providing sufficient time for the patient to recover in time for the next cycle. Many treatment regimens give a limited number of cycles, typically three to six in total. Other regimens are maintenance treatments, in which an unlimited number of cycles are given.

Hormonal treatments

Several tumour types, notably breast cancer and prostate cancer, are stimulated by endogenous hormones. Removal of this stimulus from sensitive tumours will result in their shrinkage. An understanding of the relevant endocrine pathways has identified several points for therapeutic intervention, including interference with hormone receptors and with hormone production.

Targeted treatments

The explosion of knowledge about the molecular biology of cancer has identified multiple therapeutic targets. This has ushered in an era of highly specific treatments that aim to inhibit a target essential for tumour survival while leaving other tissues unaffected. At present, most targeted therapies are not absolutely specific for their primary target and therefore do have unwanted or 'off-target' side effects. In addition, successful inhibition of the target may have inevitable undesirable consequences in addition to the desired effect, so-called 'on-target' side effects. Many targeted treatments are given in continuous cycles and the dosing and toxicity management strategies are therefore of particular importance.

These treatments will only work if a patient's tumour depends on a therapeutic target. The kinase inhibitor vemurafenib will only be effective in patients with melanoma whose

tumours have the V600E *BRAF* mutation; cetuximab is only effective in patients with colorectal cancer who have wild-type (non-mutated) *ras*; imatinib is particularly effective in patients with gastrointestinal stromal tumours who have mutations in exon 11 of the *Kit* gene – patients with mutations in exon 9 may still respond to imatinib but will require higher doses and patients without mutations in *Kit* are far less likely to respond to imatinib.

Treatment choice therefore requires a molecular analysis of the patient's tumour. It is this determination of treatment at the individual level that has led to the concept of 'personalised medicine'. Although major advances have been made, important targets such as *ras* remain elusive at present, although several *ras*-targeted molecules are in development.

Immunotherapy

Treatments to activate the immune system against cancer have a long history, stimulated by the observation that up to half of the volume of certain tumours was known to be made up of immune cells, which appeared to be inactive or dead. There is now an appreciation that multiple malignancies activate mechanisms whose normal purpose is to down-regulate the immune system after elimination of an infectious organism. These mechanisms are called T-cell checkpoints. Inhibition of these checkpoints can reactivate the immune cells and has had remarkable success in previously virtually untreatable diseases such as metastatic melanoma. This novel form of treatment is generally much better tolerated than cytotoxic chemotherapy but may result in side effects owing to the uncontrolled activation of the immune system. Side effects such as pneumonitis, colitis, adrenal failure and hypophysitis (pituitary inflammation) may be life-changing or life-threatening and are more common when using a combination of checkpoint inhibitors.

Immunotherapy is a very active field of research and it is likely that vaccines and engineered immune cells will increasingly enter treatment protocols in the coming years.

Principles of combined treatment

Non-surgical treatments are often used in combination. For example, radiotherapy and chemotherapy are often given together as an alternative to surgery, e.g. in the treatment of rectal, cervical, head and neck or brain cancers (***Table 12.5***). The rationale behind combination, as opposed to single-modality therapy, is straightforward and is somewhat analogous to that used for combined antibiotic therapy: it is a strategy designed to combat resistance. By the time of diagnosis many tumours will contain cancer cells that, through spontaneous mutation, have acquired resistance to individual modalities of treatment. Unlike antibiotic resistance, there is no need for previous exposure to the treatment. Spontaneous mutation rates are high enough to allow chance to permit the occurrence, and subsequent expansion, of clones of cells resistant to a treatment to which they have never been exposed. If only single-modality treatments were used, then the further expansion of these *de novo* resistant subclones would limit cure. The problem can be mitigated by, from the outset of treatment, combining treatment modalities.

There are three main principles upon which the choice of drugs for combination therapy is based: (i) use drugs active against the diseases in question; (ii) use drugs with distinct modes of action; (iii) use drugs with non-overlapping toxicities. By using drugs with different biological effects, for example by combining an antimetabolite with an agent that actively damages DNA, it may be possible to obtain a truly synergistic effect, i.e. where the effects of the two modalities together are superior to the additive effects of both separately. It is inadvisable to combine drugs with similar adverse effects: combining two highly myelosuppressive drugs may produce an unacceptably high risk of neutropenic sepsis. Where possible, combinations should be based upon a consideration of the toxicity profiles of the drugs concerned.

In considering the combination of radiotherapy and chemotherapy, radiation could be considered as just another drug. There is, in addition to synergy and toxicity, another factor to consider in the combination of drugs and radiation – the concept of spatial cooperation. Chemotherapy is a systemic treatment, radiotherapy is not. Radiotherapy is, however, able to reach sites, such as the central nervous system and testis, that drugs may not reach effectively. This is why, for example in patients treated primarily with chemotherapy for leukaemias, lymphomas and small cell lung cancer, prophylactic cranial irradiation may be part of the treatment protocol.

Summary box 12.4

Principles of combined treatment

- Use effective agents
- Use agents with different modes of action (synergy)
- Use agents with non-overlapping toxicities
- Consider spatial cooperation

Oncological emergencies

There are a limited number of true emergencies in oncology. Those that require immediate recognition and management are:

- **Cord compression**: rapid diagnosis and management to relieve pressure on the spinal cord is essential to obtain the best results for patients. Ideally, treatment should be instituted when cord compression is threatened rather than when it has already occurred. Management is likely to include steroids and either neurosurgery or radiotherapy.
- **Neutropenic sepsis**: rapid diagnosis and antibiotic therapy are essential. There is a strong inverse relationship between the time taken to start antibiotics and the chance of patient survival.
- **Immune side effects** including hypophysitis, adrenal failure and insulin-dependent diabetes.

In contrast, there are many urgent oncological situations that are very unpleasant for the patient or that can rapidly deteriorate if not recognised quickly. These include thrombosis, effusion, superior vena cava obstruction and pain, among many others.

Life after cancer

As early diagnosis becomes more common and treatment outcomes improve, so an increasing number of people are cured of cancer. However, the impact of a cancer diagnosis and its treatment is profound. Organisations such as UK-based Macmillan Cancer Support can provide information and support to patients facing the health, work and financial sequelae of cancer.

Symptom control and palliative care

The distinction between palliative and curative treatment is not always clear-cut and will become increasingly blurred as professional and public attitudes towards the management of cancer change. Twenty years ago cancer was perceived as a disease that was either cured or it was not; patients either lived or died. There was little appreciation that, for many patients, cancer might be a chronic disease. Nowadays, it is appreciated that many patients will have multiple different treatment options during their cancer journey. Five-year survival is not necessarily tantamount to cure. With the development of targeted therapies that regulate, rather than eradicate, cancer this state of affairs is likely to continue. The aim of treatment will be growth control rather than the extirpation of every last cancer cell. Patients will live with their cancers, perhaps for years. They will die with cancer, but not necessarily of cancer.

Patients fear the symptoms, distress and disruption associated with cancer almost as much as they fear the disease itself. Palliative treatment has as its goal the relief of symptoms. Sometimes this will involve treating the underlying problem, as with palliative radiotherapy for bone metastases; sometimes it will not. Sometimes it may be inappropriate to treat the cancer itself, but that does not imply that there is nothing more to be done – it simply means that there may be better ways to assuage the distress and discomfort caused by the tumour. Palliative medicine in the twenty-first century is about far more than optimal control of pain: its scope is wide and its impact immense (*Table 12.7*). The most important factor in the successful palliative management of a patient with cancer is early referral. Transition between curative and palliative modes of management should be seamless.

Common problems that may be effectively palliated include:

- **Cerebral metastases**: stereotactic radiosurgery for small lesions is highly effective, although limited to patients who are likely to survive long enough to benefit.
- **Effusions**: pleural and ascitic drains may control these chronic problems. In the case of pleural effusion pleurodesis may prevent reaccumulation.
- **Thrombosis**: increased coagulability and pressure on blood vessels make this a common problem in oncology.
- **Hypercalcaemia**: bisphosphonates may control the patient's calcium level and regular infusions will be necessary when the underlying tumour process is not controlled by other means.

TABLE 12.7 An outline of the domains and interventions included within palliative and supportive care.

Holistic needs assessment	Pain, anorexia, fatigue, dyspnoea, etc.	
	Treatment-related toxicity	
Symptom relief	Drugs	
	Surgery	
	Radiotherapy	
	Complementary therapies:	Acupuncture
		Homeopathy
		Aromatherapy, etc.
Psychosocial interventions	Psychological support Relaxation techniques Cognitive behavioural therapy Counselling Group therapy Music therapy Emotional support	
Physical and practical support	Physiotherapy Occupational therapy Speech therapy	
Information and knowledge	Macmillan	
	Maggie's centres	
Nutritional support	Dietary advice	
	Nutritional supplements	
Social support	Patients	
	Relatives and carers	
Financial support	Ensure uptake of entitlements	
	Grants from charities, e.g. Macmillan	
Spiritual support		

- **Fatigue**: this is often a difficult symptom, which is partly due to the tumour and partly due to its treatment. Encouraging aerobic exercise, even at a low level, can improve fatigue and also stimulate appetite.
- **Weight loss**: patients often lose their appetite and consequently lose weight. Eating little and often with food supplements as necessary may be effective in mitigating weight loss.
- **Fever**: recurrent fevers are a feature of certain tumours such as lymphoma and renal cell cancer. Tumour fever must be distinguished from infection and this can often only be done by exclusion.
- **Paraneoplastic syndromes**: these are varied and often difficult to recognise. Management of the underlying malignancy may not necessarily resolve the syndrome.

End-of-life care

End-of-life care is distinct from palliative care. Patients treated palliatively may survive for many years; end-of-life care concerns the last few months of a patient's life. Many issues, such as symptom control, are common to both palliative care

and end-of-life care but there are also problems that are specific to the sense of approaching death. These include a heightened sense of spiritual need, profound fear and the specific needs of those who are facing bereavement. The concept of a 'good death' has been embedded in many cultures over many centuries. Healthcare professionals deal with many deaths and sometimes forget that the patient who hopes for a good death has only one chance to get it right. This is why end-of-life care is worth considering in its own right and not as a mere appendage to palliative care.

Summary box 12.5

Issues at the end of life

- Appropriateness of active intervention
- Euthanasia
- Physician-assisted suicide
- Living wills
- Bereavement
- Spirituality
- Support to allow death at home
- The problem of the medicalisation of death

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